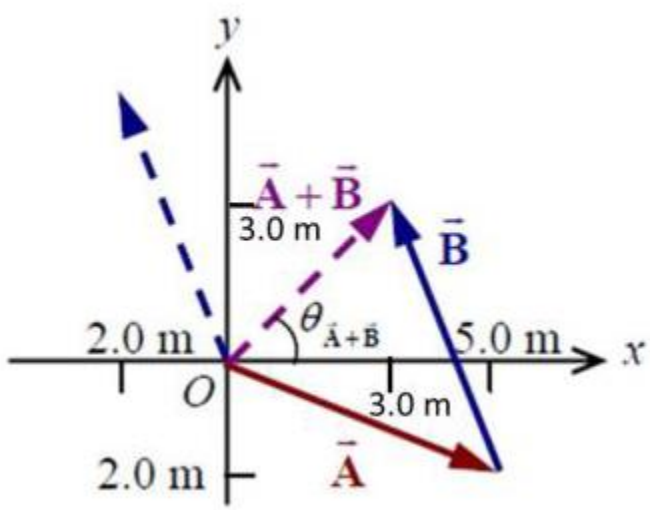
PHY101: Introduction to Physics-1

Monsoon, 2017

Tutorial-2

1. The direction cosines of a vector are the cosines of the angles it makes with the coordinate axes. The cosine of the angles between the vector and the *x*, *y*, and *z* axes are usually called, in turn α, β and γ. Prove that $\alpha^2 + \beta^2 + \gamma^2 = 1$.

2. Prove that the diagonals of an equilateral parallelogram are perpendicular.

3. Consider two points located at $\boldsymbol{r_1}$ and $\boldsymbol{r_2}$, and separated by distance $r = |\boldsymbol{r_1} - \boldsymbol{r_2}|$. Find a time-dependent vector **A(*t*)** from the origin that is at $\boldsymbol{r_1}$ at time t_1 and at $\boldsymbol{r_2}$ at time $t_2 = t_1 + T$. Assume that **A(*t*)** moves uniformly along the straight line between the two points.

4. A group of vectors are shown in the figure below.
(a). Find the direction and magnitude of ***A***
(b). Find the direction and magnitude of ***B***
(c). Find the direction and magnitude of ***A+B***